Reasoning analysis — gpt-5.4-mini (50 runs, `real_gpt54mini_med_50`)

Each model rationale is split into sentences; sentences making the same point across runs are grouped. **Shared** = the point appears in at least 60% of the runs in that group; **varying** = it appears in some runs but not most. Counts are runs, not sentences. ‘Evidence cited’ lists the numbers and times the model quoted, with the number of runs quoting each.

decision: Profile / basal 00:00

keep — 48/50 runs

Varying reasoning

- (7/48) Overnight lows exist, but they are confounded by IOB, SMB/UAM, and missing meal context.
- (6/48) Overnight lows exist, but this segment cannot be isolated from temp targets, automation, and missing meal data.
- (4/48) Overnight lows are present, but the dataset does not isolate fasting basal effect from corrections, temp targets, or residual IOB.
- (3/48) A change here would be premature without a quiet overnight verification.
- (3/48) A change would be unsupported without a clean fasting-like window.
- (3/48) No fasting-only evidence supports a basal change.
- … 16 more varying points

Points made by a single run only: 36.

Phrases used by at least half the runs: “overnight lows” (34).

Evidence cited: 00:00 (2), 02:00 (1).

change — 2/50 runs

Shared reasoning

- (2/2) The 00:00-02:00 local window shows the clearest recurrent low exposure, including severe lows in the low 40s mg/dL range.

Points made by a single run only: 3.

Phrases used by at least half the runs: “low exposure” (2), “justified changing” (1), “respects the dependency” (1), “late evening low” (1), “changing this segment” (1), “evening low exposure” (1), “meal correction” (1), “confounded by meal” (1), “highs are confounded” (1), “overnight basal change” (1), “automation effects” (1), “rule and keeps” (1).

Evidence cited: 02:00 (2), 00:00 (2).

decision: Profile / basal 05:00

keep — 50/50 runs

Varying reasoning

- (3/50) The early-morning period is not clearly failing on a fasting-like basis, and the available data do not separate basal from meal or correction effects.
- (3/50) The early morning period is not cleanly separable from overnight carryover, and the data do not isolate this segment as the cause of the low or high pattern.
- (2/50) The 03:00 to 07:00 local window is relatively stable, so there is no clean evidence for a change.

- (2/50) Early morning glucose is not clearly showing a basal-specific defect, and the observed pattern is confounded by broader overnight variability and missing meal context.
- (2/50) No fasting test is available, so a change would be speculative.
- (2/50) The morning period is mixed with rising glucose later in the day, but the data do not isolate this segment as the source of either lows or highs.
- … 11 more varying points

Points made by a single run only: 52.

Phrases used by at least half the runs: “early morning” (37).

Evidence cited: 07:00 (2), 03:00 (2), 05:00 (1), 00:00 (1).

decision: Profile / basal 06:00

keep — 50/50 runs

Varying reasoning

- (3/50) Morning glucose is not isolated from breakfast timing or automation responses, so a change is unsupported.
- (2/50) This segment may contribute to morning control, but the current dataset cannot separate it from breakfast timing, temp targets, or automation effects.
- (2/50) The morning rise begins later and could reflect meals or rising insulin needs, but the package does not separate those from basal.
- (2/50) This step may contribute to morning control, but the package does not separate basal effect from later meal and automation effects.
- (2/50) Daytime patterns are more likely to reflect meals, corrections, or dynamic sensitivity than this single basal step.
- (2/50) Morning glucose rises could reflect meals or automation compensation rather than basal insufficiency.
- … 8 more varying points

Points made by a single run only: 60.

Evidence cited: 06:00 (3), 08:00 (1).

decision: Profile / basal 10:00

keep — 50/50 runs

Varying reasoning

- (4/50) Late-morning highs exist, but they are more plausibly meal or timing related than basal-only.
- (4/50) Basal-only change is not supported.
- (4/50) Changing this segment now would be speculative.
- (3/50) The late-morning rise could reflect meals or corrections rather than basal insufficiency.
- (3/50) Late morning highs occur in the dataset, but the evidence does not isolate this basal segment because meal timing is absent and SMB/UAM are active.
- (3/50) Late-morning highs are present, but they are strongly confounded by meal timing and automation responses.
- … 16 more varying points

Points made by a single run only: 36.

Phrases used by at least half the runs: “late morning” (39).

Evidence cited: 10:00 (1).

decision: Profile / basal 16:00

keep — 50/50 runs

Varying reasoning

- (4/50) This lower afternoon basal sits in the same period as the strongest high window, but the evidence does not separate basal from meal timing or correction effects.
- (3/50) The later 18:00-19:00 lows also argue against an unverified increase.
- (3/50) This lower afternoon rate coincides with the time of highest glucose, but meals, corrections, and temp targets remain plausible explanations.
- (3/50) This is the lowest basal segment and overlaps with the highest average glucose period, but that period is also the most confounded by meals, corrections, and automation.
- (3/50) The evidence is not strong enough to change a dependent basal segment yet.
- (2/50) This lower afternoon basal coincides with the strongest daytime rise, but the same window is heavily confounded by meal effects and automated correction behavior, so a change would be premature.
- … 10 more varying points

Points made by a single run only: 53.

Evidence cited: 18:00 (3), 19:00 (3).

decision: Profile / basal 18:00

keep — 50/50 runs

Varying reasoning

- (7/50) Evening lows are a safety concern, but the evidence does not isolate this basal period from residual IOB and correction effects.
- (4/50) There is some low exposure around 18:00 to 19:00 local, but this could reflect IOB or correction carryover rather than basal excess.
- (3/50) The evening low pattern is present, but the specific contribution of this segment cannot be separated from post-correction or post-meal effects.
- (3/50) Evening lows are present, but the segment cannot be isolated from prior IOB, corrections, temporary targets, or unlogged meals.
- (3/50) The evidence does not isolate this segment as the cause.
- (3/50) Change is not supported.
- … 15 more varying points

Points made by a single run only: 37.

Phrases used by at least half the runs: “evening lows” (32).

Evidence cited: 18:00 (4), 19:00 (3), 00:00 (2).

decision: Profile / basal 19:00

keep — 50/50 runs

Varying reasoning

- (3/50) The late-evening low pattern is real, yet the package does not prove basal causality.
- (3/50) Late-evening lows argue for caution, and the data do not support a basal increase here.
- (3/50) The safest action is to keep and verify with a meal-separated window.

- (3/50) Meal, correction, and automation effects remain plausible alternatives.
- (2/50) Local 19:00 has notable below-range time, so the safer action is to keep this segment until a no-meal observation shows a reproducible basal problem.
- (2/50) A change would risk worsening lows without stronger evidence.
- … 9 more varying points

Points made by a single run only: 49.

Evidence cited: 19:00 (4).

decision: Profile / CR 00:00

keep — 50/50 runs

Varying reasoning

- (12/50) No carb entries or meal-linked response windows exist to validate a change.
- (7/50) No carb entries exist, so this CR cannot be validated.
- (6/50) There are no carb logs or meal-linked responses to validate a change.
- (5/50) Correction windows do not isolate CR effects.
- (4/50) Correction windows are mixed, so changing CR would be unsupported.
- (3/50) No meal logs exist to test this ratio, and correction responses are confounded by IOB and automation.
- … 6 more varying points

Points made by a single run only: 39.

Phrases used by at least half the runs: “meal linked” (26).

decision: Profile / CR 04:00

keep — 50/50 runs

Varying reasoning

- (5/50) No meal-linked evidence is available for this early-morning ratio.
- (4/50) Meal composition and bolus timing are unknown, so this CR segment cannot be validated.
- (4/50) There is no meal timing evidence to support a CR change, and correction responses are too confounded by automation and IOB.
- (3/50) No meal-specific evidence isolates early morning carb handling, and the observed glucose pattern does not separate CR from basal or automation effects.
- (3/50) The data do not separate meal coverage from basal or correction effects, so changing CR would be unsupported and potentially confounded.
- (3/50) CR cannot be isolated without meal logs.
- … 11 more varying points

Points made by a single run only: 38.

decision: Profile / CR 08:00

keep — 50/50 runs

Varying reasoning

- (5/50) Morning and late-morning highs may be meal-related, but the package lacks carb logs and bolus timing, so a CR change would be speculative.
- (4/50) The evidence is insufficient for a CR change.
- (4/50) Changing CR without meal evidence would be speculative.
- (3/50) Morning and midday highs may be meal related, but the package has no carb entries or bolus timing.
- (3/50) The late-morning rise could reflect meal coverage, but no carb data or bolus timing are available.
- (3/50) Keep until a logged meal shows repeatable under-coverage.
- … 8 more varying points

Points made by a single run only: 44.

Phrases used by at least half the runs: “late morning” (30).

decision: Profile / CR 16:00

keep — 50/50 runs

Varying reasoning

- (5/50) Changing CR now would be speculative.
- (4/50) Without meal logs, changing CR would be speculative and could worsen hypoglycemia risk.
- (3/50) Afternoon highs are present, but there is not enough meal-linked evidence to distinguish CR mismatch from basal, target, or activity effects.
- (3/50) A CR change would be premature.
- (3/50) Although the afternoon is the highest-glucose period, it is confounded by temp basals, corrections, and possible meals.
- (2/50) Afternoon highs are present, but the data do not separate meal coverage from basal or delayed absorption.
- … 5 more varying points

Points made by a single run only: 53.

decision: Profile / CR 20:00

keep — 50/50 runs

Varying reasoning

- (4/50) Late-evening low risk argues for caution, and there is no meal-linked evidence to support a CR change in this segment.
- (3/50) Evening control is mixed, but the lows near night and the correction variability make a CR change unsupported.
- (3/50) Keeping the CR unchanged is the safer choice until meal-linked evidence exists.
- (3/50) Evening data do not show a clean meal pattern that would justify a ratio change.
- (3/50) No meal-linked evidence isolates the evening ratio.
- (2/50) Evening lows limit the safety margin for more aggressive meal dosing, and the dataset does not provide meal-linked evidence for a CR change.
- … 10 more varying points

Points made by a single run only: 44.

decision: Profile / ISF 00:00

keep — 50/50 runs

Varying reasoning

- (7/50) Dynamic sensitivity is active, so the static ISF cannot be cleanly judged from these windows.
- (6/50) Dynamic sensitivity is already active and correction responses are mixed.
- (5/50) Correction responses vary widely and are confounded by IOB, temp targets, and likely meal timing.
- (5/50) The data do not support a reliable ISF change.
- (4/50) Correction responses are mixed and confounded by IOB, temp basals, and automation.
- (4/50) Keep because correction responses are mixed and heavily confounded by dynamic sensitivity, SMB/UAM, and IOB.
- … 14 more varying points

Points made by a single run only: 31.

Phrases used by at least half the runs: “correction responses” (30).

decision: Profile / target 00:00

keep — 50/50 runs

Varying reasoning

- (5/50) Overnight lows argue against tightening targets, and daytime highs are not isolated enough to justify a target change.
- (4/50) Overnight lows are the main safety issue, so lowering or raising the target without separating basal and IOB effects would be premature.
- (3/50) The current target is already fairly tight.
- (3/50) The target is already tight, and the main safety issue is hypoglycemia.
- (3/50) Lowering the target would increase risk without clear evidence of benefit.
- (3/50) A target change is not supported by the data.
- … 6 more varying points

Points made by a single run only: 50.

decision: Profile / target 08:00

keep — 50/50 runs

Varying reasoning

- (6/50) Keep until basal and meal timing are clearer.
- (3/50) The late-morning rise could tempt a lower target, but the data do not separate target effects from meals and automation compensation.
- (3/50) The daytime pattern is not specific enough to justify target changes, especially given nocturnal low exposure and automation confounding.
- (3/50) A target change is not supported.
- (2/50) Keep because the late-morning rise is not cleanly attributable to the target, and lowering it now could increase hypoglycemia risk in a period already showing variability.
- (2/50) The daytime highs are not clearly target-driven, and lowering or raising this target is not supported by the current evidence.
- … 10 more varying points

Points made by a single run only: 40.

decision: Profile / target 23:00

keep — 50/50 runs

Varying reasoning

- (5/50) Late-evening and overnight lows argue against making the target more aggressive.
- (5/50) Keep because late-evening and overnight lows are already a concern, so there is no support for a tighter target.
- (3/50) Overnight lows make target tightening inappropriate without cleaner evidence.
- (2/50) The late-evening lows are safety-relevant, but changing target alongside basal or automation would confound interpretation.
- (2/50) Late-evening lows argue against a lower target, but the current evidence does not isolate this segment enough to support a target change.
- (2/50) There is no clean evidence that the late-night target is responsible for the observed lows or highs.
- … 6 more varying points

Points made by a single run only: 47.

decision: Profile / DIA

keep — 50/50 runs

Varying reasoning

- (4/50) DIA cannot be validated from the available windows because insulin action is mixed with SMB/UAM, temp basals, and corrections.
- (4/50) Keep DIA unchanged until meal-linked bolus timing is available.
- (3/50) A 10-hour DIA may contribute to late insulin tail, but the available correction windows are too confounded to prove it is too long or too short.
- (3/50) The correction and automation windows do not provide clean evidence that DIA is mis-specified.
- (3/50) No evidence supports shortening or lengthening DIA at this stage.
- (3/50) The correction and IOB patterns are too confounded to justify a DIA change.
- … 9 more varying points

Points made by a single run only: 50.

Evidence cited: 10h (1).

decision: AAPS / SMB and UAM

keep — 50/50 runs

Varying reasoning

- (11/50) SMB and UAM are active and coherent with the observed loop behavior.
- (5/50) SMB and UAM appear to be functioning as intended and are likely helping manage residual rise.
- (4/50) The dataset does not support changing this before basal and meal separation is improved.
- (3/50) No material inconsistency or safety problem is shown.
- (3/50) There is no safety reason to disable them based on this dataset.
- (2/50) SMB and UAM appear coherent and useful in the observed system, and there is no material inconsistency that would justify disabling either component from this package alone.
- … 9 more varying points

Points made by a single run only: 49.

Phrases used by at least half the runs: “smb and uam” (26).

decision: AAPS / SMB activation conditions

keep — 50/50 runs

Varying reasoning

- (9/50) No material inconsistency or safety concern requires a change.
- (8/50) The activation pattern is internally coherent with the observed loop behavior.
- (8/50) The activation conditions are internally consistent and do not show a material safety conflict.
- (5/50) These settings are internally consistent and align with the observed temporary-target behavior.
- (4/50) The activation rules are internally consistent.
- (4/50) The activation logic is internally consistent and conservative.
- … 13 more varying points

Points made by a single run only: 26.

Phrases used by at least half the runs: “internally consistent” (34).

Evidence cited: 135mg/dl (2).

decision: AAPS / SMB strength and frequency

keep — 50/50 runs

Varying reasoning

- (3/50) A change is not supported without meal-linked evidence.
- (2/50) The SMB cadence is assertive but not obviously unsafe, and the observed lows argue against making it more aggressive.
- (2/50) The evidence does not support changing strength or frequency.
- (2/50) Remaining time in an SMB waiting message is not the configured SMB interval.
- (2/50) The cadence is tight, but there is no evidence that the interval or minute caps are misconfigured.
- (2/50) The delivery cadence and limits fit the observed microbolusing behavior.
- … 11 more varying points

Points made by a single run only: 53.

decision: AAPS / max basal and max IOB limits

keep — 50/50 runs

Varying reasoning

- (20/50) The observed enacted rates and IOB remain well below these ceilings, so the limits are not binding.

- (2/50) Observed temp basal and IOB values are well below these limits, so they are unlikely to explain the patterns or to be currently constraining therapy in a problematic way.
- (2/50) The limits are far above observed delivery and IOB peaks, so they are not the current bottleneck.
- (2/50) The main concern is safety from lows, not a clearly too-low ceiling.
- (2/50) The reviewed loop examples do not show these limits as the active bottleneck.
- (2/50) Excess IOB is not demonstrated as a settings problem here, so no change is supported.
- … 3 more varying points

Points made by a single run only: 52.

decision: AAPS / Autosens and Dynamic ISF

keep — 50/50 runs

Varying reasoning

- (18/50) Dynamic sensitivity is active and a sensitivity_ratio near 1 is neutral, not disabled Autosens.
- (4/50) Dynamic sensitivity is active and Autosens is intentionally off; this is not a conflict.
- (4/50) Dynamic sensitivity appears active and neutral-to-helpful.
- (3/50) Dynamic sensitivity is active even with autosens off, and sensitivity_ratio=1 is neutral rather than disabled.
- (3/50) Autosens being false does not mean sensitivity adaptation is absent, because dynamic sensitivity is active.
- (2/50) Dynamic sensitivity is already active and neutral-looking; the data do not separate sensitivity effects from meal timing or basal timing, and overnight lows make tightening riskier.
- … 14 more varying points

Points made by a single run only: 37.

Phrases used by at least half the runs: “dynamic sensitivity” (34).

decision: AAPS / carbohydrate absorption model

keep — 50/50 runs

Varying reasoning

- (25/50) There are no carb entries or e-carb records to validate absorption settings against.
- (15/50) There is no meal-linked evidence to support changing the absorption model.
- (8/50) With no meal-linked evidence, a change would be speculative.
- (3/50) The current values are not contradicted by the telemetry, and changing them would be speculative.
- (3/50) No supported change is available.
- (2/50) No carb entries or e-carb trials exist in the package, so absorption settings cannot be validated against meal-linked evidence and should not be changed first.
- … 5 more varying points

Points made by a single run only: 24.

Phrases used by at least half the runs: “carb entries” (30), “meal linked” (25).

profile recommendation

basal — 49/50 runs

Shared reasoning

- (42/49) Keep the current basal profile unchanged for now.

Varying reasoning

- (4/49) The basal profile is complete, but the observed daytime rise and overnight/evening lows are still confounded by missing meal logs, active SMB/UAM, temp targets, and correction insulin.
- (4/49) The daytime rise and late-day/overnight lows are real, but the data do not isolate basal from meal timing, corrections, temp targets, or IOB.
- (4/49) Because lows are already meaningful, changing basal now would risk worsening hypoglycemia without clean evidence.
- (3/49) The strongest pattern is overnight or early-morning lows together with late-morning to afternoon highs, but the evidence remains confounded by missing carb logs, corrections, temp targets, and dynamic sensitivity.
- (3/49) The 14-day trace shows overnight and late-evening lows plus daytime highs, but those patterns remain confounded by corrections, SMB/UAM, temp targets, site and sensor events, and missing meal logs.
- (3/49) That makes a basal change unsupported at this stage.
- … 18 more varying points

Points made by a single run only: 52.

Phrases used by at least half the runs: “profile unchanged” (40), “basal profile” (40), “basal profile unchanged” (36), “unchanged for now” (36), “now and verify” (33), “basal change” (30).

Evidence cited: 16:00 (2), 0.55u/h (2), 00:00 (2), 09:00 (1), 0.6u/h (1), 02:00 (1).

cr — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “timing on one” (1), “recurring late morning” (1), “related than proven” (1), “now and verify” (1), “proven cr defect” (1), “overnight and late” (1), “clean meal linked” (1), “revisiting cr” (1), “meal before revisiting” (1), “late evening lows” (1), “confounded to support” (1), “argue against broad” (1).

primary recommendation

basal — 34/50 runs

Shared reasoning

- (22/34) Keep the basal profile unchanged and verify with one fasting-like window.

Varying reasoning

- (7/34) The daytime rise is real, but the evidence is still too confounded by corrections, temp targets, and absent meal logs to support a basal change yet.
- (4/34) Keep the current basal profile unchanged and validate it with one clean overnight or fasting-like window plus one meal-tagged daytime window before revisiting basal or CR.
- (3/34) Keep the current basal profile unchanged and verify with one meal-free 10:00-16:00 local window before discussing any basal segment change.
- (3/34) Keep the basal profile unchanged and verify the daytime pattern first.
- (3/34) Hold basal steady and verify with a clean fasting window.
- (2/34) Keep the current basal and target profile unchanged until one fasting-style overnight observation and one late-afternoon no-meal observation can separate basal effects from meal and IOB effects.
- … 12 more varying points

Points made by a single run only: 32.

Phrases used by at least half the runs: “profile unchanged” (23), “unchanged and verify” (20).

Evidence cited: 00:00 (2), 16:00 (2), 10:00 (2), 0.6u/h (1), 0.55u/h (1), 02:00 (1).

profile — 13/50 runs

Shared reasoning

- (8/13) Keep the current profile unchanged and verify with meal-linked observation.

Varying reasoning

- (5/13) The current data do not separate basal, CR, ISF, DIA, and automation effects well enough to justify a profile change.
- (3/13) The data do not isolate a basal, CR, ISF, target, or DIA defect.
- (2/13) Keep the current profile unchanged and verify with one clean fasting-like block.
- (2/13) The data do not support changing basal, CR, ISF, target, or DIA yet because the observed highs and lows remain confounded by missing meal logs and by active automation.
- (2/13) Keep the current profile unchanged and verify the overnight low tendency versus the daytime rise with one carefully documented observation window.
- (2/13) Basal, CR, and ISF signals are all confounded by missing meal data and overlapping automation effects, while lows make tightening risky.
- … 2 more varying points

Points made by a single run only: 10.

Phrases used by at least half the runs: “basal cr” (11), “basal cr isf” (10), “profile unchanged” (9), “unchanged and verify” (7).

Evidence cited: 02:00 (1), 00:00 (1).

basal_and_targets — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “target change” (1), “confounded single basal” (1), “afternoon highs” (1), “profile unchanged” (1), “meal correction” (1), “overnight lows” (1), “keep the profile” (1), “automation effects” (1), “lows but meal” (1), “dependent edit” (1), “considering any dependent” (1), “correction and automation” (1).

meal — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “tuning is low” (1), “meal bolus behavior” (1), “declared meal strategy” (1), “targets and mixed” (1), “keep meal bolus” (1), “carb tuning” (1), “carb entries” (1), “logged with timing” (1), “delayed responses keep” (1), “bolus and carb” (1), “recurring meals” (1), “entries or declared” (1).

profile_and_data — 1/50 runs

Points made by a single run only: 3.

Phrases used by at least half the runs: “support dependent profile” (1), “one fully logged” (1), “profile unchanged” (1), “basal cr isf” (1), “safest next step” (1), “keep the profile” (1), “change the safest” (1), “validate with one” (1), “timing and cgm” (1), “enough to support” (1), “response before revisiting” (1), “one meal day” (1).

summary

50 runs

Varying reasoning

- (9/50) Good CGM coverage and a complete active profile support review.
- (7/50) Overall, the safest synthesis is to keep the current profile and automation unchanged for now and verify patterns with better meal-linked observation before considering any dependent edits.
- (6/50) The clearest pattern is recurrent overnight and late-evening hypoglycemia risk alongside late-morning to afternoon hyperglycemia, with moderate-to-high variability.
- (4/50) Over 14 days, CGM quality was good and the active AAPS profile is complete, but the dominant pattern is daytime hyperglycemia from late morning through afternoon with recurrent overnight and late-evening lows.
- (4/50) The available data do not support changing the selected profile yet; meal attribution remains limited because there are no carb treatment entries or declared meal-strategy fields.
- (4/50) CGM coverage and profile completeness are strong enough for review.
- … 14 more varying points

Points made by a single run only: 56.

Phrases used by at least half the runs: “cgm coverage” (29), “late morning” (29), “aaps profile” (25).

Evidence cited: 86.0% (4), 121.6mg/dl (2), 5.1% (1), 42mg/dl (1), 32.6% (1), 16:00 (1), 10:00 (1), 02:00 (1), 00:00 (1).

issues

50 runs

Varying reasoning

- (14/50) Recurrent hypoglycemia is the main safety concern, including lows into the low 40s mg/dL.
- (13/50) No carb treatment entries or meal timing logs are available, which blocks confident meal-bolus and e-carb interpretation.
- (10/50) A recurring late-morning to afternoon rise is visible in local-time glucose, with average values peaking around 14:00-16:00 local.
- (8/50) {‘area’: ‘safety’, ‘title’: ‘Recurrent hypoglycemia exposure’, ‘description’: ‘Below-range time is 5.1% and the data include lows down to the low-40s mg/dL, so tighter insulin changes would need caution.’} {‘area’: ‘pattern’, ‘title’: ‘Afternoon hyperglycemia window’, ‘description’: ‘Local-time averages rise through the afternoon into early evening, but this is confounded by corrections, temp targets, and automation.’} {‘area’: ‘attribution’, ‘title’: ‘No carb logs prevent meal attribution’, ‘description’: ‘Because there are no carb treatment entries, it is not possible to distinguish meal absorption from bolus timing, CR, or SMB/UAM effects.’} {‘area’: ‘interpretation’, ‘title’: ‘Mid-period profile switch limits clean comparison’, ‘description’: ‘A profile switch during the period makes same-day before-and-after comparisons less reliable.’}
- (4/50) Late morning through afternoon glucose is often elevated, with local-time averages peaking around 15:00-16:00.
- (4/50) A late-morning to afternoon glucose rise is reproducible in local-time data, but its cause is not separable from meals, corrections, temp targets, and IOB.
- … 22 more varying points

Points made by a single run only: 46.

Phrases used by at least half the runs: “late morning” (33), “low 40” (28), “treatment entries” (26).

Evidence cited: 16:00 (15), 42mg/dl (7), 5.1% (6), 17:00 (6), 32.6% (5), 02:00 (5), 13:00 (5), 6h (4), 14:00 (4), 18:00 (3), 15:00 (3), 00:00 (3).

step findings: data_quality

50 runs

Shared reasoning

- (44/50) CGM coverage is 98.7% with 3995 valid entries, no invalid or duplicate entries, and only 10 gaps with the largest gap 65 minutes.

Varying reasoning

- (14/50) The active AAPS profile is present and authoritative.
- (6/50) CGM and telemetry quality are good enough for pattern review.
- (6/50) The biggest limitation is the absence of carb treatment entries and meal annotations, which blocks clean meal attribution.
- (6/50) The authoritative selected AAPS profile is complete for basal, CR, ISF, targets, DIA, SMB/UAM, and safety limits.
- (5/50) The active profile is complete and authoritative in the package.
- (4/50) The profile context is clear because the top-level aaps_profile is authoritative.
- … 9 more varying points

Points made by a single run only: 12.

Phrases used by at least half the runs: “3995 valid” (31), “10 gaps” (31), “65 minutes” (29), “cgm coverage” (27), “largest gap” (25).

Evidence cited: 98.7% (44), 65min (29), 5min (1), 24h (1).

step findings: safety_overview

50 runs

Varying reasoning

- (22/50) Time in range was 86.0%, time below range 5.1%, time above range 8.9%, and CV was 32.6%.
- (8/50) The main safety concern is hypoglycemia risk, especially overnight and late evening.
- (6/50) Lows cluster overnight and late evening; highs cluster late morning through afternoon.
- (5/50) Local hourly lows cluster around 00:00 to 02:00 and 18:00 to 19:00, while highs cluster around 09:00 to 16:00.
- (5/50) Recurrent lows reached 41 to 42 mg/dL and highs reached 286 to 293 mg/dL.
- (5/50) Variability was moderate with CV 32.6%.
- … 13 more varying points

Points made by a single run only: 33.

Phrases used by at least half the runs: “below range” (25).

Evidence cited: 5.1% (29), 32.6% (24), 86.0% (19), 42mg/dl (18), 8.9% (14), 293mg/dl (13), 121.6mg/dl (6), 18:00 (5), 19:00 (4), 00:00 (4), 16:00 (3), 02:00 (3).

step findings: basal_targets

50 runs

Varying reasoning

- (23/50) Local-time glucose is highest roughly from 09:00 to 17:00, peaking around 15:00-16:00, while lower glucose and more hypoglycemia occur late evening and overnight, especially around 00:00-03:00 and again near 18:00-19:00.
- (6/50) Local hourly patterns suggest a lower-glucose tendency overnight and early evening, with higher glucose late morning through afternoon.
- (3/50) The afternoon rise is not cleanly basal-driven because it overlaps meals, corrections, temp targets, and automation responses.
- (3/50) This could reflect basal shape, meal timing, correction timing, temp targets, or IOB rather than a single profile defect.
- (3/50) The target schedule is already modest and does not obviously explain the pattern.

- (3/50) The target schedule itself is internally coherent.
- … 14 more varying points

Points made by a single run only: 53.

Evidence cited: 16:00 (18), 17:00 (7), 00:00 (7), 10:00 (4), 15:00 (4), 13:00 (4), 14:00 (3), 90mg/dl (3), 23:00 (3), 08:00 (3), 02:00 (3), 03:00 (3).

step findings: profile_cr_isf_dia

50 runs

Varying reasoning

- (13/50) CR varies by time segment, ISF is flat at 56 mg/dL/U, and DIA is 10 hours.
- (8/50) CR, ISF, and DIA cannot be isolated cleanly from this package.
- (7/50) Correction responses are mixed and heavily confounded by IOB, temp targets, and likely meal timing, so there is no clean basis to adjust CR, ISF, or DIA from the current data alone.
- (4/50) Correction responses are variable, including both strong drops and delayed rebounds, so a static profile conclusion is not reliable yet.
- (4/50) The data do not separate meal coverage from correction behavior well enough to justify a profile change.
- (3/50) There are no carb treatment entries, correction responses are mixed, and automation is active, so a profile change is not supported.
- … 11 more varying points

Points made by a single run only: 63.

Phrases used by at least half the runs: "cr isf" (31), "isf and dia" (27).

Evidence cited: 10h (31), 56mg/dl (21), 13.2g/u (3), 16:00 (2), 04:00 (2), 12.6g/u (2), 10.5g/u (1), 11.1g/u (1), 20:00 (1), 13g/u (1), 08:00 (1).

step findings: automation_smb_uam

50 runs

Varying reasoning

- (14/50) SMB and UAM are both enabled, SMB is always enabled, and SMB is allowed with temp targets.
- (9/50) SMB and UAM are enabled and internally consistent with the observed loop behavior.
- (8/50) SMB and UAM are both enabled and the activation rules are internally coherent.
- (7/50) SMB and UAM are active, internally consistent, and appear to be functioning as intended.
- (6/50) SMB and UAM are enabled and actively used.
- (4/50) UAM is not duplicate COB counting.
- … 9 more varying points

Points made by a single run only: 67.

Phrases used by at least half the runs: "smb and uam" (45), "uam are enabled" (27).

Evidence cited: 135mg/dl (1).

step findings: automation_sensitivity_limits

50 runs

Varying reasoning

- (27/50) Autosens is disabled but dynamic sensitivity is enabled, which is not a contradiction.
- (17/50) Max basal and max IOB limits are not binding in the observed data.
- (8/50) Max basal 12 U/h and max IOB 25 U do not appear to be limiting the observed data.
- (7/50) Sensitivity_ratio values around 1 are neutral current adjustment, not evidence of disabled Autosens.
- (7/50) A sensitivity_ratio of 1 is neutral current adjustment, not proof that sensitivity logic is off.
- (5/50) The max basal and max IOB limits are far above the observed delivery in this period and do not appear to be the main constraint.
- … 11 more varying points

Points made by a single run only: 38.

Phrases used by at least half the runs: “max basal” (38), “max iob” (38), “dynamic sensitivity” (38), “basal and max” (25).

Evidence cited: 25u (8), 12u/h (8), 70% (4).

step findings: meal_bolus_strategy

50 runs

Varying reasoning

- (12/50) No declared meal strategy fields were supplied in analysis_context, and telemetry contains no carb treatment entries.
- (7/50) Therefore no reliable prebolus timing or immediate bolus percentage can be inferred from the package.
- (6/50) The observed correction windows do not support a precise prebolus or immediate-bolus-percentage change.
- (6/50) There is no user-declared meal strategy to compare against because analysis_context.meal_strategies is empty and strategy_stable_for_period is null.
- (5/50) The safest interpretation is to keep meal handling conservative and avoid assuming that SMB/UAM should replace meal prebolus or bolus timing.
- (5/50) There is no declared meal-strategy baseline in the package, and the telemetry does not contain carb entries.
- … 10 more varying points

Points made by a single run only: 58.

Phrases used by at least half the runs: “meal strategy” (36), “declared meal” (29), “immediate bolus” (29), “declared meal strategy” (28), “meal bolus” (26), “bolus percentage” (25).

Evidence cited: 10h (1).

step findings: meal_ecarbs_absorption

50 runs

Varying reasoning

- (23/50) There are no carb treatment entries, no extended carb entries, and no e-carb durations.
- (10/50) There is no evidence for routine e-carbs in this package.
- (8/50) E-carbs should not be inferred from the aggregate response data and should not be treated as a pump extended bolus.
- (7/50) No carb treatment entries or extended carb entries are present, and the available response windows are correction windows rather than meal absorption windows.

- (5/50) There is no evidence supporting a current e-carb strategy.
- (4/50) There is therefore no supported basis to infer e-carb duration, start offset, or absorption pattern.
- … 12 more varying points

Points made by a single run only: 37.

Phrases used by at least half the runs: “extended carb” (31), “carb entries” (30), “extended carb entries” (28).

Evidence cited: 6h (2), 7h (1).

step findings: synthesis_plan

50 runs

Varying reasoning

- (17/50) The safest synthesis is to keep the current profile and automation settings unchanged for now.
- (6/50) After contradiction audit, the package supports keeping the current profile and automation unchanged.
- (5/50) The clearest supported action is to keep the profile and automation settings unchanged for now, because the evidence does not separate basal, CR, ISF, DIA, and meal timing well enough to justify dependent changes.
- (4/50) The safest coherent interpretation is: keep the current profile and automation settings, because the daytime highs and overnight/late-evening lows are both real but remain confounded by missing meal data, corrections, temporary targets, and device events.
- (3/50) After contradiction audit, the package supports a conservative plan: keep the current profile and automation settings unchanged for now, because the strongest signal is hypoglycemia risk that is not cleanly separable from IOB, corrections, temp targets, and missing meal data.
- (3/50) The main conflict in the package is mixed control rather than a single faulty setting: lows are recurrent overnight and late evening, while highs recur late morning through mid-afternoon.
- … 18 more varying points

Points made by a single run only: 72.

Phrases used by at least half the runs: “profile and automation” (29), “unchanged for now” (25).

Evidence cited: 00:00 (2), 02:00 (1).

meal strategy

50 runs

Shared reasoning

- (31/50) Keep any meal bolus approach conservative for now: do not increase the immediate bolus percentage from this dataset alone, use a modest verified prebolus when meals are logged, let SMB/UAM handle residual rise when targets and IOB allow it, and do not add routine e-carbs unless a specific slow-absorption meal pattern is later documented.

Varying reasoning

- (6/50) There is no declared meal strategy in analysis_context, no carb treatment entries, and no e-carb entries to validate a meal strategy change.
- (5/50) There are no carb treatment entries and no declared per-meal strategy values to reproduce.
- (3/50) Meal attribution is weak because carb logging is absent and the available response windows are confounded by IOB, temp basal, SMB, and UAM.
- (3/50) The package has no meal carb entries, no meal strategy declaration, and only correction-based response windows.
- (3/50) Correction responses are mixed, and the observed highs and lows are confounded by IOB, temporary targets, and automation, so there is no basis for a split or e-carb change.

- (3/50) Because lows already occur, more delayed insulin or routine e-carbs would be risky without meal-linked confirmation.
- … 15 more varying points

Points made by a single run only: 87.

Phrases used by at least half the runs: “immediate bolus” (44), “meal strategy” (43), “bolus percentage” (34), “immediate bolus percentage” (33), “declared meal strategy” (31), “meal linked” (25).

Evidence cited: 6h (4), 10h (1), 100% (1).